Practical No. 5

Aim: To detect object of interest(face) in real time and to keep tracking of the same object

Software: Jupyter/Python IDLE/PyCharm

Outcomes: Face Detection in Artificial Intelligence technology that can identify human faces in a digital image or video

Libraries:
cv2

Theory:

Face detection is the process of automatically identifying and locating human faces in digital images or video streams. It is a foundational task in computer vision, enabling applications like biometric authentication, surveillance, smartphone unlocking, and social media tagging.

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1. Definition

- Face detection is **not the same as face recognition**.
    - **Detection**: Identifies whether a face is present and its location.
    - **Recognition**: Determines whose face it is by comparing with stored data.

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2. Challenges in Face Detection

- **Variability in human faces**: Different poses, expressions, orientations, and occlusions (glasses, masks, facial hair).
- **Environmental factors**: Lighting conditions, camera resolution, and background clutter.
- **Real-time constraints**: Systems must balance accuracy with speed for live video feeds.

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3. Traditional Approaches

- **Haar Cascades (used in your code)**

    - Developed by Viola and Jones (2001).
    - Uses simple rectangular features (like edge and line detectors).
    - Applies a cascade of classifiers to quickly eliminate non-face regions.
    - Pros: Fast, lightweight, works well for frontal faces.

    - Cons: Struggles with angled faces, poor lighting, or occlusions.
- **Histogram of Oriented Gradients (HOG)**
    - Captures edge and gradient structures of faces.
    - Often combined with Support Vector Machines (SVMs).
    - More robust than Haar cascades but slower.

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4. Modern Approaches

- **Deep Learning (CNN-based models)**
    - Convolutional Neural Networks (CNNs) learn hierarchical features directly from data.
    - Examples: MTCNN (Multi-task Cascaded Convolutional Networks), YOLO (You Only Look Once), RetinaFace.
    - Pros: High accuracy, robust to variations in pose, lighting, and occlusion.
    - Cons: Computationally expensive, requires large datasets.

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5. Applications

- **Security & Surveillance**: Monitoring public spaces.
- **Authentication**: Unlocking smartphones, secure access systems.
- **Social Media**: Auto-tagging friends in photos.
- **Retail & Marketing**: Customer analytics, emotion detection. ijset.in

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📊 Comparison of Techniques

Technique	Pros	Cons	Use Case
Haar Cascades	Fast, lightweight, easy to use	Sensitive to lighting/angles	Simple webcam apps
HOG + SVM	More robust than Haar	Slower, less scalable	Academic research
CNN-based (MTCNN, YOLO, RetinaFace)	High accuracy, robust	Requires GPU, large datasets	Modern AI systems

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Code:

```
import cv2
def detectFace():

   face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades +
'haarcascade_frontalface_default.xml')

   cap = cv2.VideoCapture(0)
   if not cap.isOpened():
       print("Error: Could not open webcam.")
       return

   while True:
       ret, frame = cap.read()
       if not ret:
           print("Error: Could not read frame.")
           break

       gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

       faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5,
minSize=(30, 30))

       for (x, y, w, h) in faces:
           cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0), 2)

       cv2.imshow('Face Detection', frame)

       if cv2.waitKey(1) & 0xFF == ord('q'):
           break

   cap.release()
   cv2.destroyAllWindows()


detectFace()
```

Explanation:

Here's a clear breakdown of what each line in your code does:

Function Definition

def detectFace():

- Defines a function named `detectFace` that will handle face detection using OpenCV.

Load Haar Cascade Classifier

face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')

- Loads a pre-trained Haar Cascade classifier for detecting faces.
- `cv2.data.haarcascades` gives the path to OpenCV's built-in Haar cascade files.
- `'haarcascade_frontalface_default.xml'` is the specific file used for detecting frontal faces.

Open Webcam

cap = cv2.VideoCapture(0)

- Opens the default webcam (device index `0`).
- `cap` is the video capture object used to read frames.

```
if not cap.isOpened():
    print("Error: Could not open webcam.")
    return
```

- Checks if the webcam opened successfully.
- If not, prints an error message and exits the function.

Frame Capture Loop

while True:

- Starts an infinite loop to continuously capture frames from the webcam.

ret, frame = cap.read()

- Reads a frame from the webcam.
- `ret` is a boolean indicating success.
- `frame` is the actual image captured.

```
if not ret:
    print("Error: Could not read frame.")
    break
```

- If frame capture fails, prints an error and breaks the loop.

Convert to Grayscale

```
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
```

- Converts the captured frame from color (BGR format) to grayscale.
- Face detection works better on grayscale images.

Detect Faces

```
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(30, 30))
```

- Detects faces in the grayscale image.
- `scaleFactor=1.1`: Specifies how much the image size is reduced at each scale (controls detection accuracy).
- `minNeighbors=5`: Specifies how many neighbors each candidate rectangle should have to retain it (filters false positives).
- `minSize=(30, 30)`: Minimum size of detected faces.

Draw Rectangles Around Faces

```
for (x, y, w, h) in faces:
    cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0), 2)
```

- Loops through all detected faces.
- `(x, y, w, h)` are the coordinates and size of each face.
- Draws a green rectangle (`(0, 255, 0)`) around each face with thickness `2`.

Display the Frame

```
cv2.imshow('Face Detection', frame)
```

- Displays the frame with rectangles drawn around detected faces in a window titled **"Face Detection"**.

Exit Condition

```
if cv2.waitKey(1) & 0xFF == ord('q'):
    break
```

- Waits for a key press for 1 millisecond.
- If the user presses the **'q'** key, the loop breaks (program exits).

Release Resources

```
cap.release()
cv2.destroyAllWindows()
```

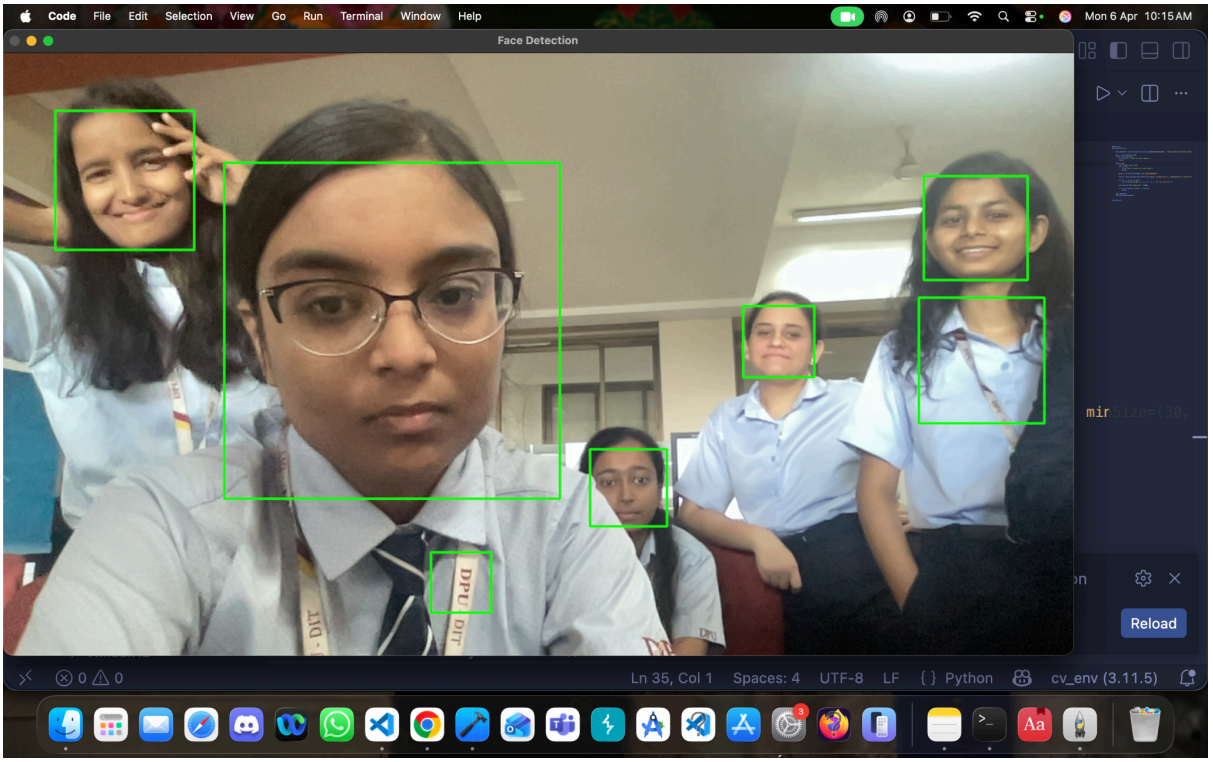
- Releases the webcam resource.
- Closes all OpenCV windows opened during execution.

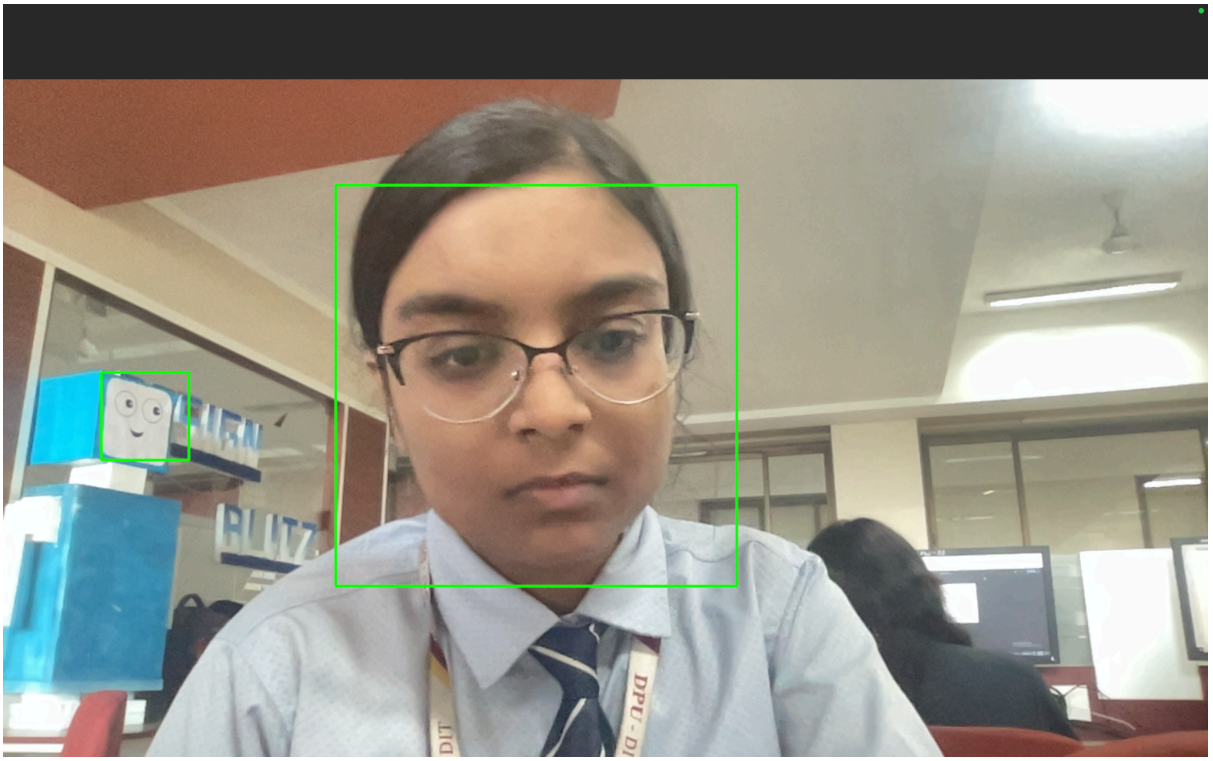
Function Call

```
detectFace()
```

- Calls the `detectFace` function to start the face detection process.

Output:


Code File Edit Selection View Go Run Terminal Window Help
Mon 6 Apr 10:15 AM
Face Detection
Reload
Ln 35, Col 1 Spaces: 4 UTF-8 LF { } Python cv_env (3.11.5)



Conclusion:

Face detection is a **fundamental computer vision task** that enables machines to locate human faces in images or video streams. Over time, techniques have evolved from **traditional methods** like Haar cascades and HOG descriptors to **modern deep learning approaches** using convolutional neural networks.

- **Traditional methods** (e.g., Haar cascades) are lightweight and fast, making them suitable for simple real-time applications, but they struggle with variations in pose, lighting, and occlusion.
- **Deep learning methods** (e.g., MTCNN, YOLO, RetinaFace) provide far greater accuracy and robustness, though they require more computational power and large datasets.
- The choice of technique depends on the **application requirements**: speed vs. accuracy vs. available resources.

In essence, face detection has become a **cornerstone of AI-powered systems**, enabling applications in security, authentication, entertainment, and human-computer interaction. As technology advances, detection systems are becoming increasingly reliable, adaptable, and integrated into everyday life.

Would you like me to also give a **visual diagram of the face detection pipeline** (from image input → preprocessing → detection → output), so you can see the workflow at a glance?